



西安交通大学
XI'AN JIAOTONG UNIVERSITY

第六章

Alternating Direction Method of Multipliers (ADMM)

来源: Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers (Boyd, Parikh, Chu, Peleato, Eckstein)

目录

- 大规模优化问题
 - ✓ 面向大规模数据的机器学习/统计
 - ✓ 大规模网络的动态优化
- 分散优化（Decentralized optimization）
 - ✓ 多设备/多处理器等协调求解大规模问题

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now
the essence of knowledge

Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers

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Borja Peleato⁴ and Jonathan Eckstein⁵



对偶问题

$$\begin{array}{ll}\min & f(x) \\ \text{s.t.} & Ax = b\end{array}$$

- 函数 $f: \mathbb{R}^n \rightarrow \mathbb{R}$ 为二次连续可微严格凸函数;
- $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = m < n$;
- $p^* = \inf \{f_0(x): Ax = b, f_i(x) \leq 0, i = 1, 2, \dots, m\} = f(x^*)$

- ✓ Lagrangian函数 $L: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$, $L(x, y) = f(x) + y^\top (Ax - b)$
- ✓ 对偶函数: $g(y) = \inf_x L(x, y) = -f^*(-A^\top y) - b^\top y$
- ✓ 对偶问题: $\max_y g(y)$, 记 $y^* = \operatorname{argmax}_y g(y)$
- ✓ 由对偶问题最优解求得原问题最优解: $x^* = \operatorname{argmin}_x L(x, y^*)$



对偶上升方法

$$\begin{array}{ll} \min & f(x) \\ \text{s.t} & Ax = b \end{array}$$

- Lagrangian函数 $L: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$, $L(x, y) = f(x) + y^\top (Ax - b)$
- $\nabla g(y)$ 的计算: 假设对偶函数 $g(y) = \inf_x L(x, y) = -f^*(-A^\top y) - b^\top y$ 可微, 记

$x^+ = \operatorname{argmin}_x L(x, y)$, 有

$$\nabla g(y) = Ax^+ - b$$

- 对偶上升方法——合适的 α^k 使得对偶函数值逐步增大

$$x^{k+1} := \operatorname{argmin}_x L(x, y^k)$$

$$y^{k+1} := y^k + \alpha^k (Ax^{k+1} - b)$$



x 最小化

对偶更新, 步长 $\alpha^k > 0$



- 合适的假设和 α^k 取值能够保证 $x^k \rightarrow x^*$, $y^k \rightarrow y^*$, 然而实际中很难满足



对偶分解

- 假设目标函数 $f(x)$ 是可分的, 即 $\forall x = (x_1, x_2, \dots, x_N)$, $x_i \in \mathbb{R}^{n_i}$, $f(x) = \sum_{i=1}^N f_i(x_i)$, 相应的矩阵 A 分块为 $A = [A_1 \ A_2 \ \dots \ A_n]$, 使得 $Ax = \sum_{i=1}^N A_i x_i$, 则

$$\begin{array}{ll} \min_x f(x) \\ \text{s.t. } Ax = b \end{array} \quad \longrightarrow \quad \begin{array}{ll} \min_{x_1 \in \mathbb{R}^{n_1}, \dots, x_n \in \mathbb{R}^{n_N}} \sum_{i=1}^N f_i(x_i) \\ \text{s.t. } \sum_{i=1}^N A_i x_i = b \end{array}$$

- Lagrangian函数 $L(x, y)$ 是可分的, 即

$$L(x, y) = \sum_{i=1}^N L_i(x_i, y) = \sum_{i=1}^N \left(f_i(x_i) + y^\top A_i x_i - \frac{1}{N} y^\top b \right)$$

- 对偶上升方法

$$x_i^{k+1} := \operatorname{argmin}_{x_i} L_i(x_i, y^k)$$

$$y^{k+1} := y^k + \alpha^k \left(\sum_{i=1}^N A_i x_i^{k+1} - b \right)$$



x_i 并行最小化

对偶更新, 步长 $\alpha^k > 0$



增广拉格朗日(Augmented Lagrangian)及乘子方法

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & Ax = b \end{array} \quad \longleftrightarrow \quad \begin{array}{ll} \min & f(x) + \frac{\rho}{2} \|Ax - b\|_2^2 \\ \text{s.t.} & Ax = b \end{array}$$

- 增广拉格朗日 $L_\rho: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$, $\forall \rho > 0$ (惩罚参数)

$$L_\rho(x, y) = f(x) + y^\top (Ax - b) + \frac{\rho}{2} \|Ax - b\|_2^2$$

✓ $\rho > 0$ 为惩罚参数, $\rho = 0$ 时, $L_0(x, y) = L(x, y)$

- 相应的对偶问题 $g_\rho(y) = \inf_x L_\rho(x, y)$

- 乘子方法

$$\begin{array}{l} x^{k+1} := \operatorname{argmin}_x L_\rho(x, y^k) \\ y^{k+1} := y^k + \rho(Ax^{k+1} - b) \end{array} \quad \begin{array}{c} \uparrow x \text{ 最小化} \\ \downarrow \text{对偶更新} \end{array}$$

对偶上升方法

$$\begin{array}{l} x^{k+1} := \operatorname{argmin}_x L(x, y^k) \\ y^{k+1} := y^k + \alpha^k (Ax^{k+1} - b) \end{array}$$



增广拉格朗日(Augmented Lagrangian)及乘子方法

$$\min f(x)$$

$$\text{s.t. } Ax = b$$



$$Ax^* - b = 0$$

$$\nabla f(x^*) + A^T y^* = 0$$

- 增广拉格朗日 $L_\rho(x, y) = f(x) + y^T(Ax - b) + \frac{\rho}{2} \|Ax - b\|_2^2$

- 乘子方法

$$x^{k+1} := \operatorname{argmin}_x L_\rho(x, y^k)$$

↑ x 最小化

$$y^{k+1} := y^k + \rho(Ax^{k+1} - b)$$

↓ 对偶更新



(x^{k+1}, y^{k+1}) 是对偶可行的, 因为

$$0 = \nabla_x L_\rho(x^{k+1}, y^k)$$

$$= \nabla_x f(x^{k+1}) + A^T (y^k + \rho(Ax^{k+1} - b))$$

$$= \nabla_x f(x^{k+1}) + A^T y^{k+1}$$

- 当 $Ax^{k+1} - b \rightarrow 0$ 时, 原问题可行性取得, 故乘子方法具有较好的收敛性质
- 即使函数 $f(x)$ 可分, $L_\rho(x, y)$ 不可分, 乘子方法中 x 最小化步不可并行计算

Alternating Direction Method of Multipliers (ADMM)

- 融合对偶上升方法可分性和乘子方法良好收敛性的ADMM，求解以下形式优化问题

$$\begin{aligned} \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad & f(x) + g(z) \\ \text{s.t.} \quad & Ax + Bz = c \end{aligned}$$

- ✓ 两个变量，可分目标函数
- ✓ 函数 $f(x), g(z)$ 为凸函数
- ✓ 矩阵 $A \in \mathbb{R}^{p \times n}$, $B \in \mathbb{R}^{p \times m}$, $c \in \mathbb{R}^p$
- ✓ 最优值 $p^* = \inf\{f(x) + g(z) \mid Ax + Bz = c\}$



ADMM

$$\begin{aligned} \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad & f(x) + g(z) \\ \text{s.t.} \quad & Ax + Bz = c \end{aligned}$$

- 增广的 $L_\rho: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, $\forall \rho > 0$ (惩罚参数),

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top (Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|_2^2$$

- ADMM算法

$$x^{k+1} := \operatorname{argmin}_x L_\rho(x, z^k, y^k)$$

$$z^{k+1} := \operatorname{argmin}_z L_\rho(x^{k+1}, z, y^k)$$

$$y^{k+1} := y^k + \rho(Ax^{k+1} + Bz^{k+1} - c)$$

↑ x 最小化
 z 最小化
↓ 对偶更新

- 对比乘子方法

$$(x^{k+1}, z^{k+1}) := \operatorname{argmin}_{x, z} L_\rho(x, z, y^k)$$

$$y^{k+1} := y^k + \rho(Ax^{k+1} + Bz^{k+1} - c)$$



最优性条件 & 停止准则

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top (Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|_2^2$$

$$x^{k+1} := \operatorname{argmin}_x L_\rho(x, z^k, y^k)$$

$$z^{k+1} := \operatorname{argmin}_z L_\rho(x^{k+1}, z, y^k)$$

$$y^{k+1} := y^k + \rho(Ax^{k+1} + Bz^{k+1} - c)$$

- 最优性条件 $\left\{ \begin{array}{l} \text{原问题可行: } Ax^* + Bz^* - c = 0 \\ \text{对偶问题可行: } \begin{cases} 0 \in \partial f(x^*) + A^\top y^* & (1) \\ 0 \in \partial g(z^*) + B^\top y^* & (2) \end{cases} \end{array} \right. \quad \partial \text{为子梯度算子}$

- z^{k+1}, y^{k+1} 总满足对偶可行性条件(2), 因为
$$0 \in \partial g(z^{k+1}) + B^\top y^k + \rho B^\top (Ax^{k+1} + Bz^{k+1} - c) = \partial g(z^{k+1}) + B^\top y^{k+1}$$

- 原问题残差 $r^{k+1} = Ax^{k+1} + Bz^{k+1} - c$;

- 对偶问题残差 $s^{k+1} = \rho A^\top B(z^{k+1} - z^k)$:
$$0 \in \partial f(x^{k+1}) + A^\top y^k + \rho A^\top (Ax^{k+1} + Bz^k - c) = \partial f(x^{k+1}) + A^\top y^{k+1} + \rho A^\top B(z^k - z^{k+1})$$

$$s^{k+1} = \rho A^\top B(z^{k+1} - z^k) \in \partial f(x^{k+1}) + A^\top y^{k+1}$$

ADMM变形

$$\begin{aligned} \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad & f(x) + g(z) \\ \text{s.t.} \quad & Ax + Bz = c \end{aligned}$$

- 增广的 $L_\rho: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, $\forall \rho > 0$ (惩罚参数),

$$\begin{aligned} L_\rho(x, z, y) &= f(x) + g(z) + y^\top (Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|_2^2 \\ &= f(x) + g(z) + \frac{\rho}{2} \|Ax + Bz - c + u\|_2^2 + \text{const} \end{aligned}$$

$$u = \frac{1}{\rho} y^k$$

- ADMM算法变形

$\left\{ \begin{array}{l} \uparrow \\ x \text{ 最小化} \\ z \text{ 最小化} \\ \downarrow \\ \text{对偶更新} \end{array} \right.$	$x^{k+1} := \operatorname{argmin}_x f(x) + \frac{\rho}{2} \ Ax + Bz^k - c + u^k\ _2^2$
	$z^{k+1} := \operatorname{argmin}_z g(z) + \frac{\rho}{2} \ Ax^{k+1} + Bz - c + u^k\ _2^2$
	$u^{k+1} := u^k + (Ax^{k+1} + Bz^{k+1} - c)$

- 原残差: $r^k = Ax^k + Bz^k - c$, 则 $u^k = u^0 + \sum_{j=1}^k r^j$



最优性条件 & 停止准则

$$\begin{aligned}x^{k+1} &:= \operatorname{argmin}_x f(x) + \frac{\rho}{2} \|Ax + Bz^k - c + u^k\|_2^2 \\z^{k+1} &:= \operatorname{argmin}_z g(z) + \frac{\rho}{2} \|Ax^{k+1} + Bz - c + u^k\|_2^2 \\u^{k+1} &:= u^k + (Ax^{k+1} + Bz^{k+1} - c)\end{aligned}$$

- 原问题残差 $r^{k+1} = Ax^{k+1} + Bz^{k+1} - c$
- 对偶问题残差 $s^{k+1} = \rho A^\top B(z^{k+1} - z^k)$
- 目标函数逼近残差

$$f(x^k) + g(z^k) - p^* \leq - (y^k)^\top \mathbf{r}^k + (x^k - x^*)^\top \mathbf{s}^k$$

- 算法停止准则

$$\begin{cases} \|r^k\|_2 \leq \epsilon^{pri} \\ \|s^k\|_2 \leq \epsilon^{dual} \end{cases}$$

ADMM收敛性

$$\begin{aligned} \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad & f(x) + g(z) \\ \text{s.t.} \quad & Ax + Bz = c \end{aligned}$$

- 假设条件：1. 函数 $f: \mathbb{R}^n \rightarrow \mathbb{R}$ 及 $g: \mathbb{R}^m \rightarrow \mathbb{R}$ 满足 $\text{epi } f$ 及 $\text{epi } g$ 为非空闭凸集合

$$\text{epi } f = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}$$

$$\text{epi } g = \{(x, t) \in \mathbb{R}^m \times \mathbb{R} \mid g(x) \leq t\}$$

- 2. Lagrangian函数 $L_0(x, z, y)$ 有鞍点，即存在 (x^*, z^*, y^*) 使得 $\forall x, y, z$ 满足

$$L_0(x^*, z^*, y) \leq L_0(x^*, z^*, y^*) \leq L_0(x, z, y^*)$$

- 以上假设条件满足时，ADMM迭代满足
 - ✓ 原残差收敛： $k \rightarrow \infty$ 时，原残差 $r^k = Ax^k + Bz^k - c \rightarrow 0$
 - ✓ 目标函数收敛： $k \rightarrow \infty$ 时，目标函数 $f(x^k) + g(z^k) \rightarrow p^*$
 - ✓ 对偶变量收敛： $k \rightarrow \infty$ 时，对偶变量 $y \rightarrow y^*$ (对偶最优解)



有限制的凸优化问题

$$\begin{array}{ccc} \min_x f(x) & \xrightarrow{\substack{g \text{ 为凸集合 } \mathcal{C} \text{ 的指示函数} \\ g(x) = \begin{cases} 0, & x \in \mathcal{C} \\ +\infty, & \text{otherwise} \end{cases}}} & \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^n} f(x) + g(z) \\ \text{s.t. } x \in \mathcal{C} & & \text{s.t. } x - z = 0 \end{array}$$

• ADMM算法

$$\begin{array}{l} \left. \begin{array}{l} \uparrow x \text{ 最小化} \\ z \text{ 最小化} \\ \downarrow \text{对偶更新} \end{array} \right\} \begin{array}{l} x^{k+1} := \operatorname{argmin}_x f(x) + \frac{\rho}{2} \|x - z^k + u^k\|_2^2 \\ z^{k+1} := \Pi_{\mathcal{C}}(x^{k+1} + u^k) \quad \leftarrow \text{集合 } \mathcal{C} \text{ 上的 Euclidean 投影} \\ u^{k+1} := u^k + x^{k+1} - z^{k+1} \end{array} \end{array}$$

• 算法停止准则

$$\left\{ \begin{array}{l} \|r^k\|_2 = \|x^k - z^k\|_2 \leq \epsilon^{pri} \\ \|s^k\|_2 = \|-\rho(z^k - z^{k-1})\|_2 \leq \epsilon^{dual} \end{array} \right.$$

LASSO

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1 \xrightarrow{\text{ADMM}} \begin{array}{l} \min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|z\|_1 \\ \text{s.t. } x - z = 0 \end{array}$$

• ADMM算法

$\uparrow$ x 最小化
 $\downarrow$ z 最小化
 对偶更新

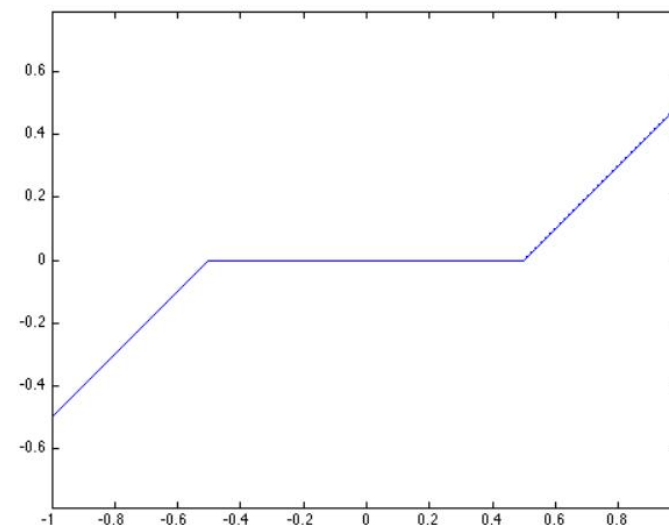
$$x^{k+1} := (A^T A + \rho I)^{-1} (A^T b + \rho(z^k - u^k))$$

$$z^{k+1} := S_{\lambda/\rho}(x^{k+1} + u^k)$$

$$u^{k+1} := u^k + x^{k+1} - z^{k+1}$$

这里，实数软阈值算子 $S_\kappa(\alpha) = \begin{cases} \alpha - \kappa, & \alpha > \kappa \\ 0, & |\alpha| \leq \kappa \\ \alpha + \kappa, & \alpha < -\kappa \end{cases}$

$$= \left(1 - \frac{\kappa}{|a|}\right)_+ a$$



GROUP LASSO

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \sum_{i=1}^N \|x_i\|_2 \xrightarrow{\text{ADMM}} \begin{aligned} &\min_{x,z} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \sum_{i=1}^N \|z_i\|_2 \\ &\text{s.t. } x_i - z_i = 0 \ (i = 1, 2, \dots, N) \end{aligned}$$

这里, $x = (x_1, x_2, \dots, x_N)$, $x_i \in \mathbb{R}^{n_i}$, $z = (z_1, z_2, \dots, z_N)$, $z_i \in \mathbb{R}^{n_i}$

• ADMM算法

x 最小化	$x^{k+1} := (A^\top A + \rho I)^{-1} (A^\top b + \rho(z^k - u^k))$
z 最小化	$z_i^{k+1} := S_{\frac{\lambda}{\rho}}(x_i^{k+1} + u^k) \ (i = 1, 2, \dots, N)$
对偶更新	$u^{k+1} := u^k + x^{k+1} - z^{k+1}$

这里, 向量软阈值算子 $S_\kappa(\alpha) = \left(1 - \frac{\kappa}{\|a\|_2}\right)_+ a$

Consensus

- 含有N个目标项的多线程优化问题

$$\min_{z \in \mathbb{R}^n} f(z) = \sum_{i=1}^N f_i(z) \quad \xrightarrow[\text{局部变量 } x_i \in \mathbb{R}^n]{\text{ADMM}}$$

$$\begin{aligned} \min_{z, x_i} \quad & \sum_{i=1}^N f_i(x_i) \\ \text{s.t.} \quad & x_i - z = 0 \quad (i = 1, 2, \dots, N) \end{aligned}$$

- 全局一致性问题

- 局部变量 $x_i \in \mathbb{R}^n$
- 全局变量 $z \in \mathbb{R}^n$
- 一致性 $x_i - z = 0$

ADMM算法	↑ x 最小化	$x_i^{k+1} := \operatorname{argmin}_{x_i} \left(f_i(x_i) + (y_i^k)^\top (x_i - z^k) + \frac{\rho}{2} \ x_i - z^k\ _2^2 \right)$
	z 最小化	$z^{k+1} := \frac{1}{N} \sum_{i=1}^N \left(x_i^{k+1} + \frac{1}{\rho} y_i^k \right)$
	对偶更新 ↓	$y_i^{k+1} := y_i^k + \rho (x_i^{k+1} - z^{k+1})$

Consensus with Regularization

- 含有N个目标项的多线程优化问题

$$\min_{z \in \mathbb{R}^n} f(x) = \sum_{i=1}^N f_i(z)$$

ADMM
局部变量 $x_i \in \mathbb{R}^n$

$$\begin{aligned} \min_{z, x_i} \quad & \sum_{i=1}^N f_i(x_i) \\ \text{s.t.} \quad & x_i - z = 0 \quad (i = 1, 2, \dots, N) \end{aligned}$$

$$g(z) = \|z\|_1$$

$$\begin{aligned} \min_{z, x_i} \quad & \sum_{i=1}^N f_i(x_i) + g(z) \\ \text{s.t.} \quad & x_i - z = 0 \quad (i = 1, 2, \dots, N) \end{aligned}$$

- ADMM算法

x最小化

z最小化

对偶更新

$$\begin{aligned} x_i^{k+1} &:= \operatorname{argmin}_{x_i} \left(f_i(x_i) + (y_i^k)^\top (x_i - z^k) + \frac{\rho}{2} \|x_i - z^k\|_2^2 \right) \\ z^{k+1} &:= \operatorname{argmin}_z \left(g(z) + \sum_{i=1}^N \left(- (y_i^k)^\top z + \frac{\rho}{2} \|x_i^{k+1} - z\|_2^2 \right) \right) \\ y_i^{k+1} &:= y_i^k + \rho (x_i^{k+1} - z^{k+1}) \end{aligned}$$



Consensus Classification——SVM example

- 数据: $\{(x_i, y_i) | x_i \in \mathbb{R}^n, y_i \in \{-1, 1\}\}$
- 线性分类器 $\text{sign}(W^\top x + b)$
- Loss function $l(y_i(W^\top x_i + b))$, 例如Hinge loss $l(u) = (1 - u)_+$

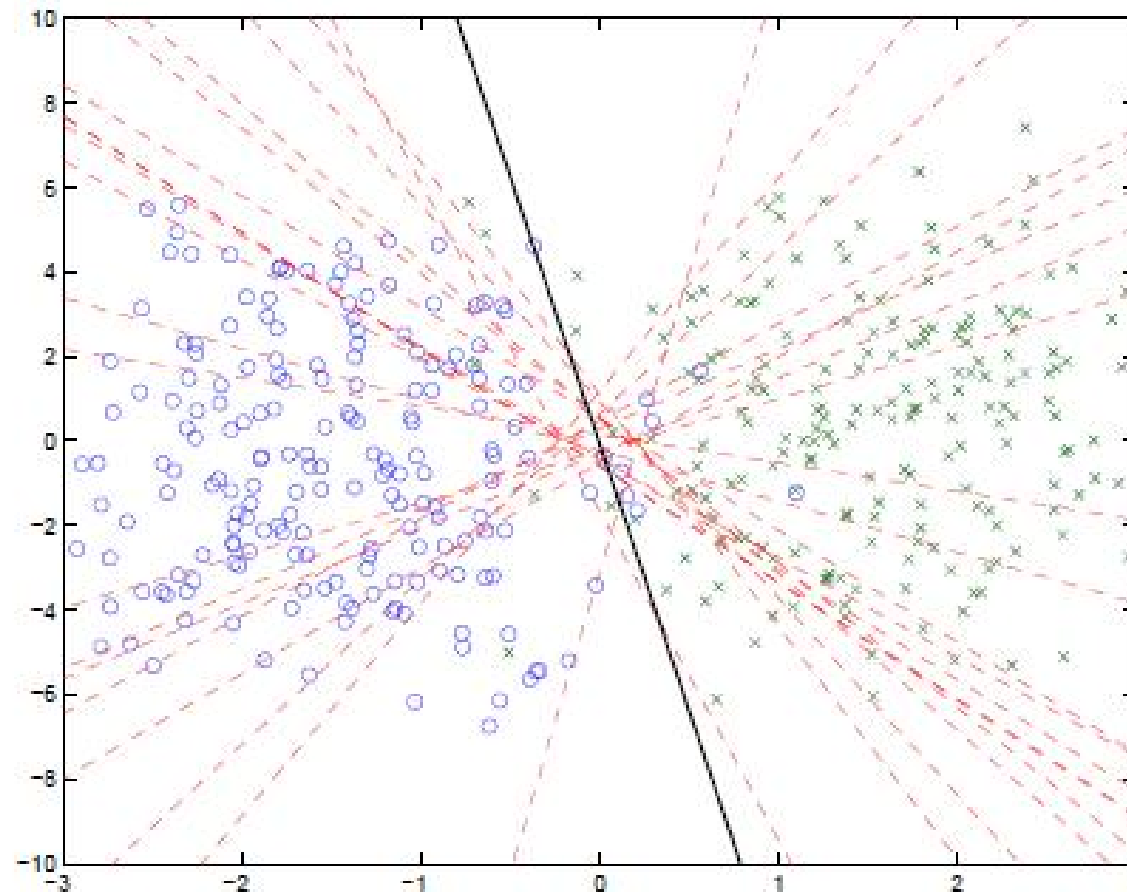
$$\min_{W, b} \frac{1}{N} \sum_{i=1}^N l(y_i(W^\top x_i + b)) + \text{Reg}(W)$$

- 分割数据集合为20个子集, 使用基于ADMM的Consensus求解



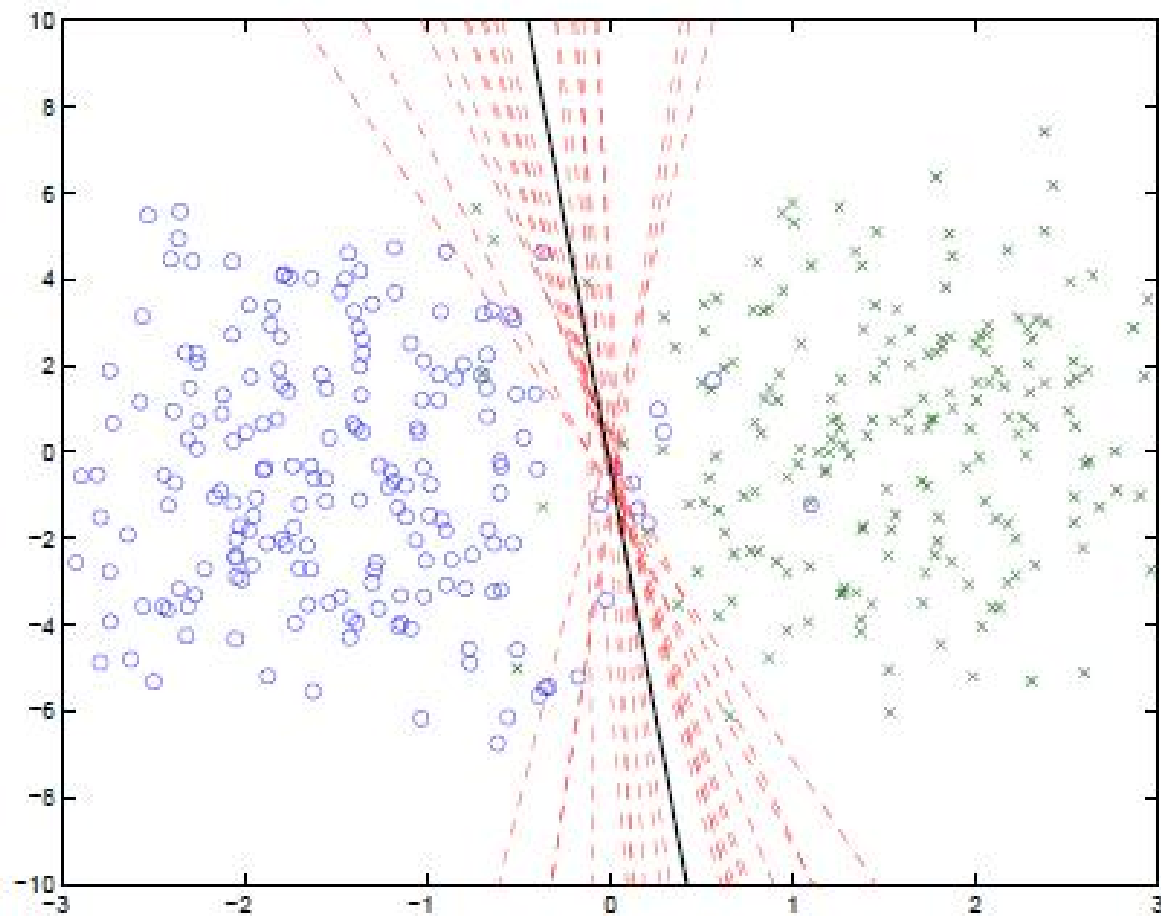
Consensus Classification—SVM example

Iteration 1



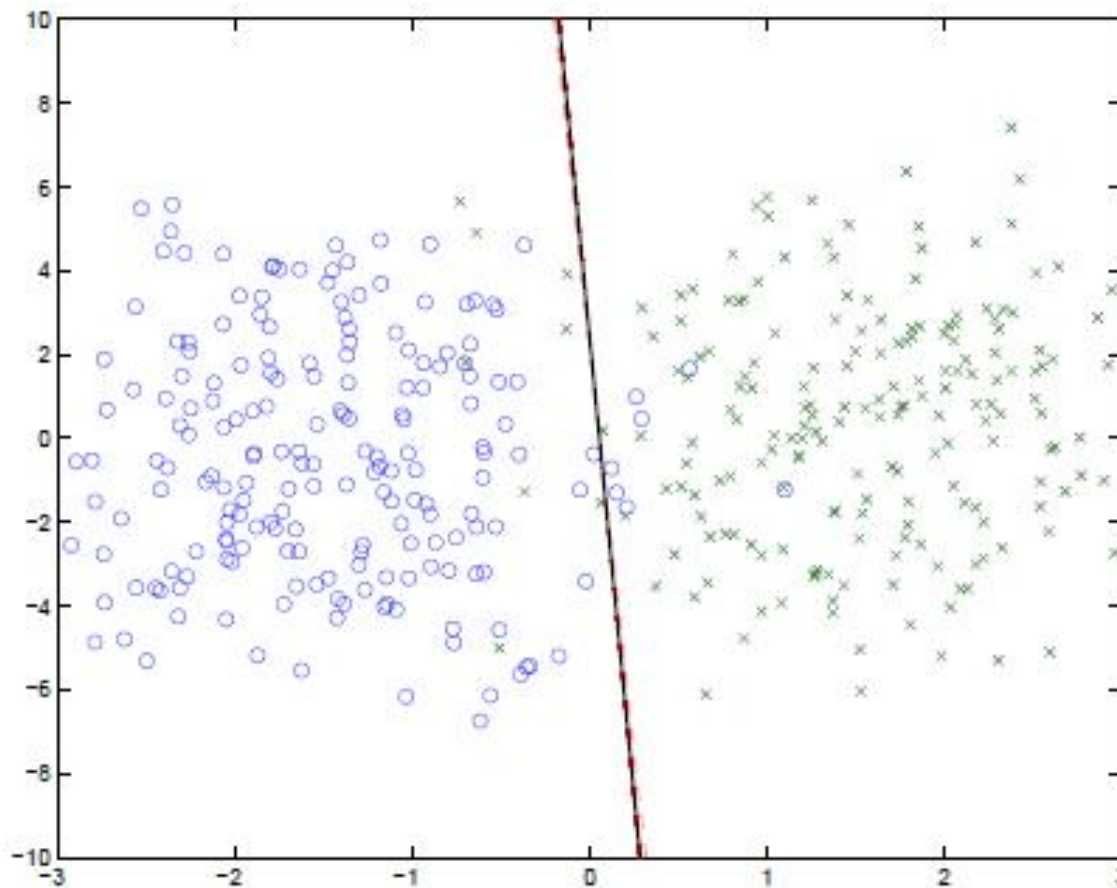
Consensus Classification—SVM example

Iteration 5



Consensus Classification—SVM example

Iteration 40



Sharing

$$\min_{x_i \in \mathbb{R}^n} \sum_{i=1}^N f_i(x_i) + g\left(\sum_{i=1}^N x_i\right)$$

- ADMM算法
- ↑
x最小化
z最小化
对偶更新
↓

ADMM
局部变量 $x_i \in \mathbb{R}^n$

$$\begin{aligned} \min_{z_i, x_i} & \sum_{i=1}^N f_i(x_i) + g\left(\sum_{i=1}^N z_i\right) \\ \text{s.t.} & x_i - z_i = 0 \quad (i = 1, 2, \dots, N) \end{aligned}$$

$$\begin{aligned} x_i^{k+1} &:= \operatorname{argmin}_{x_i} \left(f_i(x_i) + \frac{\rho}{2} \|x_i - z_i^k + u_i^k\|_2^2 \right) \\ z^{k+1} &:= \operatorname{argmin}_z \left(g\left(\sum_{i=1}^N z_i\right) + \frac{\rho}{2} \sum_{i=1}^N \|z_i - u_i^k - x_i^{k+1}\|_2^2 \right) \\ u_i^{k+1} &:= u_i^k + x_i^{k+1} - z_i^{k+1} \end{aligned}$$

总结

$$\begin{aligned} \min_{x \in \mathbb{R}^n, z \in \mathbb{R}^m} \quad & f(x) + g(z) \\ \text{s.t} \quad & Ax + Bz = c \end{aligned}$$

- 增广的 $L_\rho: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, $\forall \rho > 0$ (惩罚参数),

$$L_\rho(x, z, y) = f(x) + g(z) + y^\top (Ax + Bz - c) + \frac{\rho}{2} \|Ax + Bz - c\|_2^2$$

- ADMM算法
- $$\begin{aligned} x^{k+1} &:= \operatorname{argmin}_x L_\rho(x, z^k, y^k) \\ z^{k+1} &:= \operatorname{argmin}_z L_\rho(x^{k+1}, z, y^k) \\ y^{k+1} &:= y^k + \rho(Ax^{k+1} + Bz^{k+1} - c) \end{aligned}$$

↑

x 最小化

z 最小化

对偶更新

↓